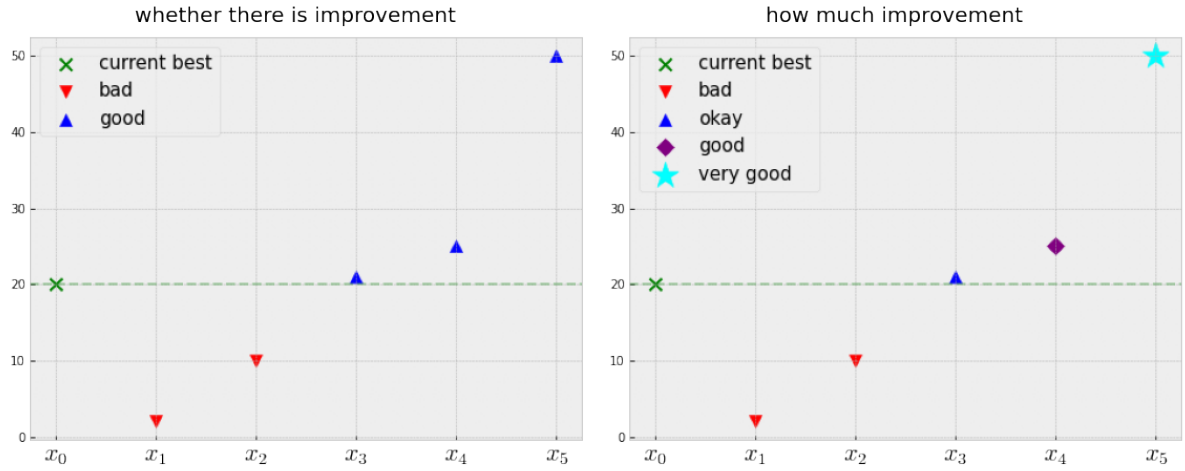


Figure 4.15 The difference between Probability of Improvement (left) and Expected Improvement (right). The former only cares whether we improve from the incumbent or not, while the latter considers how much improvement is made.

Of course, this doesn't mean that we now can simply design the policy that picks out the data point that offers the largest improvement from the incumbent, as we won't know how much (if any) an improvement we will observe until we actually query the objective function.

NOTE **Despite not knowing the exact value of the improvement we will observe, we can reason about the magnitude of the improvement of each candidate point in a probabilistic manner.**

Recall that in Figure 4.9 and Figure 4.10, we have a truncated normal distribution representing the improvement that we will observe at a given point. By computing the area of the highlighted region, we obtain the probability that a point will improve from the incumbent, giving us the Probability of Improvement policy. However, we can perform other computations as well.

IMPORTANT **Definition of the Expected Improvement policy**

In addition to the probability of improvement, we may compute the expected value of the random variable corresponding to the highlighted region. The fact that we are dealing with truncated normal distributions enables us to compute this expected value in closed form. The Bayesian optimization policy that scores data points using this measure is called Expected Improvement (EI).

While the closed-form formula for the Expected Improvement score isn't as simple as the CDF like PoI, EI's scoring formula is still just as easy to calculate. Intuitively, using the expected value of improvement may allow us to better balance between exploration and exploitation than

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